

3D Personal Portfolio Website

Asif Iftekhhar Fahim

190104027

Ahsanullah University of Science and Technology

Ahmadul Karim Chowdhury

190104037

Ahsanullah University of Science and Technology

Abstract—In the evolving landscape of modern web design, the integration of 3D graphics and animations has become a pivotal element in captivating and engaging users. This research paper presents the development and implementation of a 3D personal portfolio website using the Three.js framework, with the aim of showcasing individual skills, activities, and projects in an immersive and visually appealing manner.

The proposed methodology involves a comprehensive approach to conceptualization, asset creation, website development, user interaction, dynamic animations, customization, and information presentation. Leveraging the Three.js library, the project realizes an interactive 3D environment with intuitive controls, enabling users to explore and interact with 3D models using various input methods. Notably, the integration of dynamic animations and seamless transitions enhances the overall visual appeal and user engagement.

Furthermore, the website's responsive design ensures adaptability across different devices and screen sizes. Users are empowered to personalize the appearance of 3D models, providing a sense of ownership and alignment with personal preferences or branding. The inclusion of an interactive book element adds an innovative dimension, allowing users to access comprehensive project details and personal information.

Acknowledging limitations such as potential performance constraints, browser compatibility, and accessibility considerations, this research underscores the transformative potential of Three.js in redefining digital self-presentation. By combining JavaScript, Three.js, and Blender-created 3D assets, the resulting 3D personal portfolio website offers an exceptional user experience, emphasizing the synergy between technological innovation and creative expression.

I. INTRODUCTION

In the dynamic landscape of modern web design, static web pages have given way to interactive experiences that captivate users. The integration of 3D graphics and animations has become increasingly vital in creating visually appealing and engaging web content. As web designers and developers strive to push the boundaries of traditional web design, the adoption of powerful tools becomes essential. One such tool that empowers the creation of stunning 3D visualizations directly within web browsers is Three.js.

A cross-browser JavaScript framework and API called Three.js makes use of WebGL, a low-level graphics API created especially for the web. Three.js provides a platform-independent method for creating intricate 3D animations as a fundamental component of a website by utilizing the JavaScript language.

The motivation behind choosing Three.js lies in its transformative potential and relevance in the current web design landscape. As websites increasingly seek to captivate and engage

users, the use of 3D graphics and animations has become a necessity. Three.js provides a powerful and accessible framework for achieving this goal, enabling designers and developers to create immersive and visually striking experiences.

II. RELATED/EXISTING WORK

In the field of 3D personal websites using Three.js, several notable works have emerged. The following examples highlight the achievements and advancements made by designers and developers in this domain.

One example of a 3D personal website created using Three.js is the portfolio of designer and developer, Bruno Arizio [1]. Bruno Arizio's website features a 3D model of his head that follows the user's mouse movements. This creates an engaging and interactive experience for the user. Additionally, the website includes a 3D model of Bruno's workspace that the user can explore.

Another example is the portfolio of designer and developer, Michael Bromley [2]. However, Bromley's website takes things a step further by incorporating a 3D model of his entire body. This creates a more immersive experience for the user and allows them to interact with Bromley in a more personal way.

A third example is the portfolio of designer and developer, Tobias van Schneider [3]. However, what sets Tobias's website apart is the use of sound. The website includes ambient sound effects that create a more immersive experience for the user. This demonstrates the potential of Three.js for creating not just visually stunning websites, but also websites that engage the user's other senses.

These examples demonstrate the potential of Three.js for creating engaging and interactive 3D personal websites.

III. PROPOSED METHODOLOGY

The proposed methodology focused on utilizing Three.js to develop a 3D personal portfolio website. The goal was to create an immersive and visually appealing web experience that showcased the individual's skills, activities, and personal projects.

The 3D portfolio website, developed using Three.js, featured the following key components:

1) Conceptualization and Planning:

- Defined the project's scope, objectives, and target audience.
- Created a detailed plan for the layout, placement of 3D objects, and user interface elements.

2) Asset Creation and Design:

- Utilized Blender to design and create intricate 3D assets, ensuring optimal quality and performance for web display.
- Designed interactive elements such as the book, monitor screen, glass, and other objects, integrating them seamlessly into the scene.

3) Website Development:

- Constructed the foundational HTML structure of the website, encompassing the layout for the 3D environment and user interface components.
- Utilized CSS to implement responsive design and stylish visual elements for both dark and white modes.
- Employed JavaScript and Three.js to establish the 3D environment and interactions.

4) User Interaction and Navigation:

- Implemented intuitive controls that allowed users to interact with the 3D models through various input methods, including mouse, touch, and keyboard.
- Enabled rotation, zooming, and panning of 3D objects to enhance the user's exploration experience.
- Designed and integrated UI elements such as the PC, glass, mobile device, and navigation buttons for easy access to different sections.

5) Dynamic Animations and Transitions:

- Integrated dynamic animations and smooth transitions to create visually appealing and engaging interactions.
- Developed movement and transformation animations for individual 3D objects and seamless scene transitions.

6) Customization and Personalization:

- Provided users with the ability to personalize the appearance of 3D models, including adjusting colors, textures, and materials to align with personal preferences or branding.

7) Information Presentation:

- Designed an interactive book element that, when clicked, zoomed in and opened to display detailed personal information and project descriptions.
- Populated the book's pages with comprehensive content, including personal info, project narratives, employed techniques, and external links.

8) Integration with Social Media:

- Linked relevant social media accounts (Facebook, YouTube, LinkedIn) within the "Get in Touch" section, allowing users to easily connect and follow.

9) Responsive Design Implementation:

- Ensured that the website seamlessly adapted to various screen sizes and devices, optimizing the user experience across desktop, tablet, and mobile platforms.

10) Testing and Optimization:

- Conducted thorough testing across different browsers and devices to identify and resolve compatibility issues.
- Gathered user feedback to refine the website's performance, interactions, and visual aspects.

The result was an engaging and interactive 3D personal portfolio website that effectively showcased the individual's achievements, skills, and creativity. The combination of Three.js, JavaScript, and 3D assets crafted in Blender led to a seamless user experience, captivating visual design, and an innovative approach to digital self-presentation.

IV. LIMITATIONS

Despite the successful implementation of the proposed 3D personal portfolio website using Three.js, certain limitations need to be acknowledged:

- 1) **Performance Constraints:** The intricate 3D models and dynamic animations might lead to increased resource consumption, potentially affecting the website's loading times, especially on devices with lower processing power.
- 2) **Browser Compatibility:** While efforts were made to ensure compatibility across various browsers, certain older or less common browsers might not fully support the advanced features of the Three.js library, leading to a degraded user experience.
- 3) **Mobile Optimization:** Despite responsive design principles, the interactive nature of 3D elements might present challenges in providing a seamless experience on certain mobile devices or smaller screens.
- 4) **User Learning Curve:** The innovative user interactions and 3D exploration techniques might require users to familiarize themselves with new navigation methods, potentially impacting the user experience for those less accustomed to such interfaces.
- 5) **Content Accessibility:** The focus on visual elements might pose challenges for visually impaired users or those relying on screen readers, possibly limiting the inclusivity of the website.
- 6) **Maintenance Complexity:** Regular maintenance and updates might require specialized knowledge in 3D graphics, JavaScript, and the Three.js library, potentially making ongoing management more complex.

V. CONCLUSION

The development and implementation of the 3D personal portfolio website using Three.js have demonstrated the potential to create an engaging and interactive digital platform for showcasing personal skills and projects. The inclusion of 3D models, dynamic animations, and user interactions enhances the visual appeal and encourages deeper user engagement.

However, it is important to acknowledge the limitations outlined above. While the project successfully achieved its goals, further research and development could address the

identified constraints, potentially leading to a more refined and inclusive user experience.

In conclusion, the utilization of Three.js in building the 3D personal portfolio website has opened avenues for innovative digital self-presentation, paving the way for future advancements in interactive web design.

REFERENCES

- [1] Romainavalle. (n.d.-b). Studio—BA®. Studio-ba. <https://brunoarizio.com/>
- [2] Bromley, M. (n.d.). Michael Bromley. <https://michaelbromley.co.uk/>
- [3] Tobias van Schneider - Design & Creative Direction. (2022, May 4). House of Van Schneider. <https://vanschneider.com/>